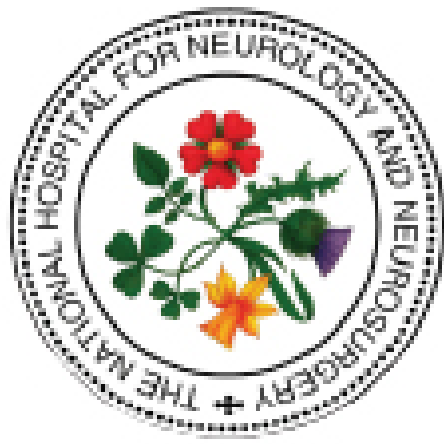




The National Hospital for Neurology and Neurosurgery

Neuroscience Critical Care Unit (NCCU)

ACUTE ISCHAEMIC STROKE (AIS)

AIS is characterised by sudden loss of blood circulation to an area of brain, typically in a vascular territory, resulting in corresponding loss of neurologic function. Ischaemic stroke is caused by thrombotic or embolic occlusion of a cerebral artery and more common than haemorrhagic stroke. Aim of treatment is to achieve recanalization and reperfusion of ischaemic penumbra with IV thrombolysis and / or endovascular thrombectomy in appropriate patients.

This protocol outlines safe targets based on agreed guidelines, which may be adapted in individual clinical circumstances with the judgement / knowledge of the responsible senior clinician

DEFINITION

AIS occurs due to a variety of underlying pathological processes and about 85% of all strokes are due to ischaemia.

Most hemispheric infarctions reach maximal swelling after 3 to 5 days. Occlusion of middle cerebral artery (MCA) or one of its branches is one of most common causes of AIS. Space occupying oedema is leading cause of mortality in 1st week following stroke and clinical features usually manifest between 2nd and 4th day. However, neurological deterioration may occur more rapidly within 2h of symptom onset, and when associated with involvement of the whole MCA territory has been termed 'malignant MCA syndrome'.

Haemorrhagic transformation: is a common complication in patients with AIS. It occurs when peripheral blood extravasates across a disrupted blood brain barrier (BBB) into the brain following ischemic stroke – factors linked to increased risk include reperfusion therapy (thrombolysis / thrombectomy), hypertension.

CURRENT AIS GUIDELINES

<https://www.strokeguideline.org/app/uploads/2023/04/National-Clinical-Guideline-for-Stroke-2023.pdf>

<https://doi.org/10.1161/STR.0000000000000211>

IMAGING

1. Non-contrast brain CT – to exclude haemorrhagic stroke
2. MRI once patient stabilised (ideally within first 24 hours)

TREATMENT

Acute management aim is to prevent infarction and minimise permanent brain damage. Early treatment is critical to rescue potentially salvageable tissue ('time is brain') with effective arterial recanalization to restore blood flow.

Patients who present <4.5 hours after stroke onset and have no contraindications may be considered for thrombolysis (see below). Patients who are not eligible for thrombolysis (due to delayed time to presentation or contraindications to thrombolysis) can be considered for endovascular therapy (see below).

BP management: Lowering BP may reduce risk of haemorrhagic transformation and oedema, whereas high BP may protect brain by maintaining cerebral perfusion when autoregulation impaired. BP targets reflect risk of worsening ischaemia versus risk of haemorrhage – patient specific BP decisions also based on imaging.

Thrombolysis: IV injection of recombinant tissue plasminogen activator (r-tPA) is gold standard treatment for ischaemic stroke, given within 4.5 hours following onset of symptoms if agreed criteria met (extended to 4.5-9 hours for 'wake-up' stroke).

Mechanical Thrombectomy (MT): intra-arterial thrombolysis or mechanical clot retrieval in eligible patients 6-16h after stroke.

Decompressive hemicraniectomy: for malignant MCA infarct should be considered (within 48 hours of symptom onset) for people with acute stroke who meet agreed criteria.

